

Welcome to the realm of Glaciers!

The world's highest peak, Mount Everest (8848 m) on the left, feeds the Khumbu Icefall and then the Khumbu Glacier which flows beneath Nuptse (7861 m) towards the right.



The base of the Khumbu Icefall with debris and frozen ponds. Everest Base Camp is on the far left.



The pitted surface of Lhotse Glacier with mounds enclosing a supraglacial pond.



The middle reach of the glacier, looking towards the snout. Note the prominent trimline and collapsing Little Ice Age moraines.



A telephoto shot of Lhotse Shar Glacier illustrating the high degree of surface-roughness, arising from uneven melting beneath the mobile debris cover.



This breach in the left-lateral moraine of Khumbu Glacier, is the main discharge route for surface drainage on the glacier.



A frozen pond in the middle of Khumbu Glacier, which here has near-complete debris cover, including massive boulders. Pumori is in the background.

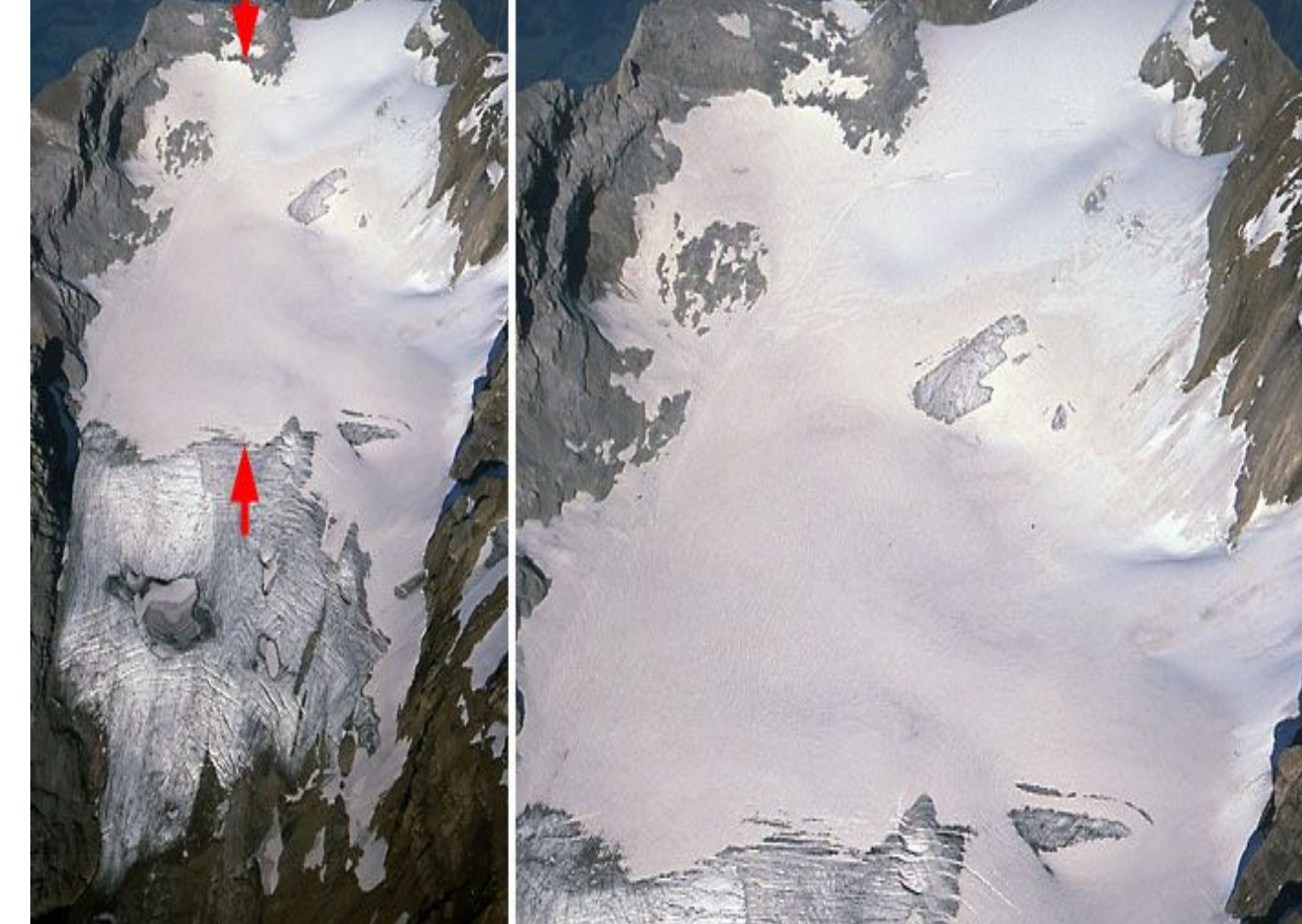


Ice cliffs, up to ten metres high, indicate that the debris cover over the whole glacier is little more than a couple of metres thick. The internal dark layers are from rock avalanches, which provide most of the debris cover.

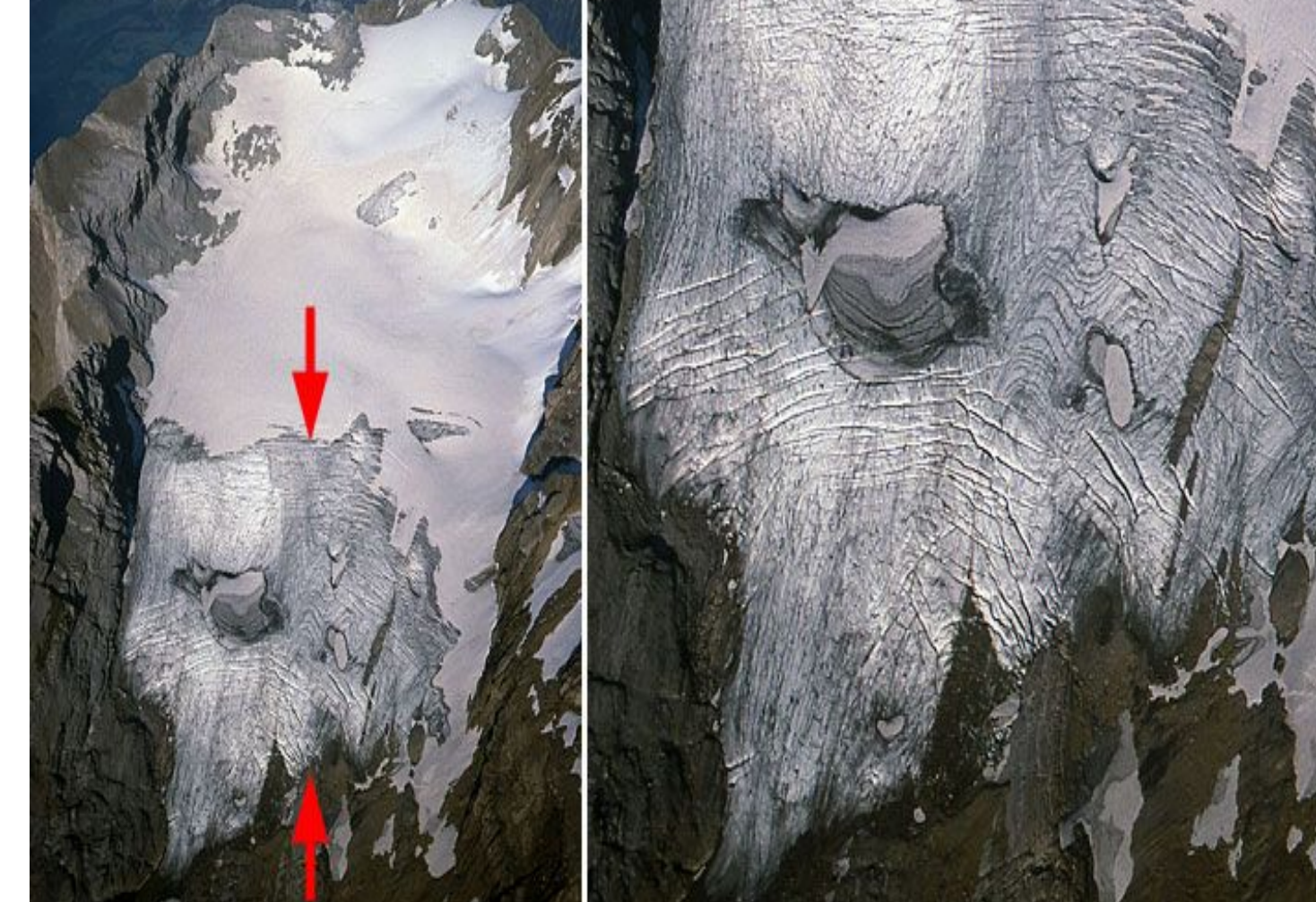


How do scientists study a glacier??

Accumulation area of Glärnischgletscher, a small mountain glacier in NE Switzerland. Aerial photo J. Alean, 1982.



Ablation area of Glärnischgletscher, a small mountain glacier in northeastern Switzerland. Aerial photo J. Alean, 1982.



Extremely rapid recession of Triftgletscher, Bernese Alps, Switzerland. Left: 2002, right: 2003. Recession is accelerated by calving in the proglacial lake. Photos J. Alean.

Retreating glaciers and climate change?

Aerial view of crevasses in the accumulation area of Fox Glacier, South Island, New Zealand. Photo M. J. Hambrey.



Cluster of cryoconite holes on Vadret da Morteratsch, Grisons, Switzerland. Photo J. Alean.



Two major flow units, Vadret Pers (1) and Vadret da Morteratsch (2) combine to form the tongue of a major valley glacier in Grisons, Switzerland. Aerial photo J. Alean, 1982.

